

# Taylor's Formula

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**Abstract.** Write your abstract here. State the problem, the contribution, and the main result in 4–8 sentences.

**Keywords:** keyword1, keyword2

## 1 Theorem

**Theorem 1.1 (Taylor's Formula).** Let  $f \in C^n([a, x])$  for some  $n \geq 1$ . Then

$$f(x) = \sum_{k=0}^{n-1} \frac{f^{(k)}(a)}{k!} (x-a)^k + R_n(x)$$

where

$$R_n(x) = R_n(x, f, a) = \int_a^x \frac{(x-t)^{n-1}}{(n-1)!} f^{(n)}(t) dt$$

*Proof.* For  $n \geq 2$

$$\begin{aligned} f(x) &= f(a) + \int_a^x f'(t) dt \\ &= f(a) + \int_a^x f'(t) d(t+C) \\ &= f(a) + [f'(t)(t+C)]_a^x - \int_a^x (t+C) f''(t) dt \end{aligned}$$

Let  $C = -x$ :

$$\begin{aligned} f(x) &= f(a) + f'(t)(x-x) - f'(t)(a-x) + \int_a^x (x-t) f''(t) dt \\ &= f(a) + f'(t)(x-a) + \int_a^x (x-t) f''(t) dt \end{aligned}$$

Assume that for all  $n \leq k$ , we have that

$$f(x) = \sum_{m=0}^{n-1} \frac{f^{(m)}(a)}{m!} (x-a)^m + \int_a^x \frac{(x-t)^{n-1}}{(n-1)!} f^{(n)}(t) dt$$

Suppose  $f \in C^{k+1}([a, b]) \subset C^k([a, b])$ : □

## 2 Mean Value Theorem for integrals

**Proposition 2.1 (Mean Value Theorem for Integrals).** *Let  $f, g \in C([a, b]; \mathbb{R})$  such that  $g \geq 0$  or  $g \leq 0$ . Then  $\exists \xi \in (a, b)$  such that*

$$\int_a^b f(x)g(x) \, dx = f(\xi) \int_a^b g(x) \, dx$$

*Proof.* Without loss of generality  $f$  not constant,  $g \geq 0$ , and  $g \not\equiv 0$ . Let

$$M = \sup_{x \in [a, b]} f(x)$$

$$m = \inf_{x \in [a, b]} f(x)$$

Thus  $m \leq f(x) \leq M \Rightarrow mg(x) \leq f(x)g(x) \leq Mg(x)$ . Thus

$$m \int_a^b g(x) \, dx \leq \int_a^b f(x)g(x) \, dx \leq M \int_a^b g(x) \, dx.$$

We get

$$\int_a^b f(x)g(x) \, dx = \lambda \int_a^b g(x) \, dx$$

Thus  $\lambda \in [m, M]$ . For closed interval we are done by IVT, but we want open interval.

1.  $m < \lambda < M$ :  $\exists c, d \in [a, b]$  such that  $m = f(c)$  and  $M = f(d)$ . By IVT, applied to  $[c, d]$  we can find some  $\xi \in (c, d)$  such that  $\lambda = f(\xi)$
2.  $\lambda = m$  or  $\lambda = M$ : Without loss of generality suppose  $\lambda = M$ . Thus

$$0 = \int_a^b (M - f(x))g(x) \, dx = 0 \Rightarrow (M - f(x))g(x) = 0 \text{ a.e.} \Rightarrow M - f(x) = 0 \text{ a.e.}$$

Thus  $f(x) \equiv M$  by continuity. □

## 3 Remainder Theorem

**Theorem 3.1 (Taylor's Remainder Formula).** *Suppose  $f \in C([a, b]; \mathbb{R})$*

*Proof.*

$$R_n(x) = \int_a^x \frac{(x-t)^{n-1}}{(n-1)!} f^{(n)}(t) \, dt$$

Applying Proposition 3.1 we see that  $\exists \xi \in (a, x)$  such that

$$\begin{aligned} R_n(x) &= f^n(\xi) \int_a^x \frac{(x-t)^{n-1}}{(n-1)!} dt \\ &= f^{(n)}(\xi) \frac{(x-t)^n}{n!} (-1) \Big|_a^x \\ &= f^{(n)}(\xi) \frac{(x-a)^n}{n!} \end{aligned}$$

□

**Example 3.1.** Let  $f \in C^2(\mathbb{R})$  be such that  $f(x) \geq 0$  for all  $x \in \mathbb{R}$  where

$$\sup(f'')_\infty < \infty.$$

$\forall x \in \mathbb{R}$  we have that

$$|f'(x)| \leq \sqrt{2\|f''\|_\infty f(x)}.$$

This is a sharp estimate. For example when  $f(x) = x^2$ . Then  $|f'(x)| = 2|x|$  and

$$\|f\|_\infty = 2.$$

Thus

$$\sqrt{2\|f''\|_\infty f(x)} = \sqrt{4x^2} = 2|x| = |f'(x)|$$

*Proof.*

$$0 \leq f(x+y) = f(x) + f'(x)y + \frac{y^2}{2}f''(\xi)$$

where  $\xi \in (x, x+y) \cup (x+y, x)$ .

$$f(x+y) \leq f(x) + f'(x)y + \frac{\|f''\|_\infty}{2}y^2$$

If  $\|f''\|_\infty \leq 0$ , thus

$$0 \leq y^2 + \frac{2f'(x)}{\|f''\|_\infty} + \frac{2f(x)}{\|f''\|_\infty} = \left(y + \frac{f'(x)}{\|f''\|_\infty}\right)^2 + \frac{2f(x)}{\|f''\|_\infty} - \frac{(f'(x))^2}{\|f''\|_\infty^2}$$

□

## 4 Real Analytic Functions

**Definition 4.1 (Real Analytic).** Let  $\Omega \subset \mathbb{R}$  be a open set. We say  $f \in C^\infty(\mathbb{R})$  is **real analytic** if for all compact  $K \subset \Omega$  such that  $C_K$  such that

$$\sup_{x \in K} |f^{(j)}(x)| \leq C^{j+1}, \text{ for all } j = 0, 1, \dots$$

*Proof.* Let  $x_0 \in \Omega$ . There exists  $r > 0$  such that  $[x - r, x + r] \subset \Omega$ . Taylor's Formula gives us that

$$f(x) = \sum_{k=0}^{n-1} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + R_n(x)$$

for  $|x - x_0| \leq r$ .

$$\begin{aligned} |R_n(x)| &\leq \left| \int_{x_0}^x \frac{f^{(n)}(t)}{(n-1)!} (x-t)^{n-1} dt \right| \\ &= C^{n+1} |x - x_0|^n. \end{aligned}$$

Thus  $\forall x_0 \in \Omega$  there exists  $V \subseteq \Omega$  a neighborhood of  $x_0$ ,

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$$

for all  $x \in V$  with absolute convergence. □

## References